

ANDrate toolbox: A Tutorial for collecting behavioral data in naturalistic stimuli in a python based toolbox

Eric Rui-Wen Wang^a, Rui Jing^a and Haiyan Wu^{a,*}

^aCentre for Cognitive and Brain Sciences and Department of Psychology, University of Macau, Taipa, 999078, Macau, China

ARTICLE INFO

Keywords:

Research software
Naturalistic Experiment
Behavioral coding
Web-based Eye Tracking
Continuous measurement
Event Segmentation

ABSTRACT

Naturalistic stimuli, such as movies, can provide ecological validity and examine human perception and cognition in real-world scenarios. As a result, it is drawing more and more attention from the psychological and cognitive neuroscience community. However, one drawback is that these types of studies usually have no explicit behaviour measures and require raters to provide group-level consensus annotations for the stimuli, such as consensus boundaries and emotional ratings. Here, we introduce *ANDrate*. *ANDrate* is an open-source annotation toolbox; it implements boundary segmentation, event annotation, and continuous two-dimensional measurement reporting. The toolbox also supports webcam recording capabilities to ensure data quality. Finally, *ANDrate* implements algorithms for calculating consensus boundaries and ratings, and enables visualization and the export of standard data formats. It is particularly well-suited for applications in naturalistic experiments, such as research providing continuous emotional states and event cognition for films.

1. Introduction

The usage of naturalistic stimuli, such as movies, has become increasingly prevalent within the fields of psychological science and cognitive neuroscience (Xu, Liu, Hu and Wu, 2025; Zheng, Chen and Wu, 2025; Wei and Wang, 2026; Zhang, Seak, Dolan and Wu, 2025; Liu, Jiang, Li and Ding, 2025). As researchers strive for greater ecological validity, experimental paradigms are shifting from static, trial-based tasks toward dynamic, naturalistic interactions. This shift offers substantial advantages, including enhanced generalizability and superior test-retest reliability (Zhang, Kim, Brang and Liu, 2021; Nastase, Goldstein and Hasson, 2020). Indeed, there has been a persistent call within the academic community to transition traditional behavioural experiments into more naturalistic ones (Nastase et al., 2020; Wang and Wu, 2026; Wang, Luo, Wang, Zhang and Wu, 2026b). However, the naturalistic approach introduces novel challenges for behavioural data collection. Due to the absence of explicit behavioural tasks, these experiments often require additional annotators to label the stimuli, ultimately calculating consensus labels at group level to serve as behavioural data. This process includes, for instance, emotional ratings (Ke, Song, Bai, Rosenberg and Leong, 2025), attentional engagement (Song, Finn and Rosenberg, 2021a), and event boundary detection (Baldassano, Chen, Zadbood, Pillow, Hasson and Norman, 2017). Within the framework of the naturalistic paradigm, these annotations are essential for investigating a broad range of psychological phenomena, such as memory, event cognition, aging, and language comprehension.

Compared to other fields of naturalistic experimentation, such as neuroimaging analysis Yang, Hu, Li, Guo, Gao, Liu, Wang, Qu, Li, Li, Li, Huang, Chen, Wen and Yuan (2025), and face-viewing Zhou, Chen, Ning, Cai, Zhao, Zhao and Moulson (2026), there are currently few open-source toolkit for annotating behavior during naturalistic stimuli. To date, existing options are largely restricted to continuous rating toolboxes, such as DARMA and CARMA, whereas the event boundary marking has been largely overlooked (Girard and Wright, 2018; Girard, 2014). Existing video annotation software lack event boundary marking capabilities, focusing solely on continuous variable tracking. Moreover, existing toolboxes do not provide accessible, user-friendly post-rating analysis, such as the calculation of consensus boundaries and consensus scores.

* Corresponding author: Haiyan Wu

Centre for Cognitive and Brain Sciences and Department of Psychology, University of Macau, Taipa, Macau, China

Address: N21-1015d, University of Macau, Taipa, Macau

haiyanwu@um.edu.mo (H. Wu)

ORCID(s): 0000-0001-8869-6636 (H. Wu)

To bridge these methodological gaps, we introduce **ANDrate**, an open-source, web-based toolbox specifically engineered for high-precision, multi-dimensional video annotation. Designed for naturalistic media annotation, **ANDrate** integrates the two primary behavioral annotation methods used in naturalistic experiments: multi-dimensional continuous rating and event boundary segmentation. In addition, it incorporates webcam-driven eye-tracking as a method for data quality control and validation. The toolbox also features post-rating analysis and visualization modules, allowing experimenters to calculate common consensus metrics, and export data for further analysis. Importantly, we have designed the toolbox with a Graphical User Interface (GUI) to maximize accessibility for the research community, while also providing open-source code to facilitate future extensions by researchers.

In this tutorial, we provide a step-by-step guide to collecting behavioral data with **ANDrate** (Figure 1). Firstly, we outline the fundamentals of naturalistic experiments, including brief introductions to continuous media rating and event boundaries. Subsequently, we demonstrate how to configure the GUI and deploy behavioral experiments for data collection. For researchers interested in developing further functionality based on the toolbox, we include an introduction of the code for each feature. We then illustrate how to use the toolbox for post-rating analysis, such as calculating consensus scores, and how to visualize the resulting data. Finally, to assist researchers new to the naturalistic paradigm, we demonstrate the entire workflow described above using a movie dataset from prior research conducted with 14 participants.

2. Common Behavior metrics in Naturalistic Experiments

Broadly speaking, the behavioral metrics commonly employed within naturalistic paradigms in psychology and cognitive neuroscience include continuous ratings (Isik and Vessel, 2019; Yang, Milisav, Kopal, Holmes, Mitsis, Misic, Finn and Bzdok, 2023) and event boundary segmentations (Chen, Leong, Honey, Yong, Norman and Hasson, 2017; Baldassano et al., 2017). These represent the majority of behavioral indices required for modern naturalistic designs utilizing media (e.g., film, audio). The core objective of these metrics is to establish a behavioral "ground truth" for the stimuli, thereby providing the necessary context for interpreting recorded neural signals. In the following sections, we briefly review the history, theory, typical experiment procedure, and considerations for collecting continuous ratings and detecting event boundaries.

2.1. Continuous behavior rating

To comprehensively understand how behaviors and experiential processes unfold dynamically over time, continuous behavioral rating methods were developed (Ruef and Levenson, 2007). Sometime this is also referred to as continuous experience sampling (Conner, Tennen, Fleeson and Barrett, 2009; Christensen, Barrett, Bliss-Moreau et al., 2003). Unlike traditional self-report methods, which are often limited to retrospective evaluations for the overall stimuli, continuous ratings capture the continuity and dynamics of psychological processes by requiring participants to report their perceptions or experiences of a stimulus at high frequencies. This could range from second-to-second (Wallace, Mckeown, Goodall-Halliwell, Chitiz, Forest, Karapanagiotidis, Mulholland, Turnbull, Vanderwal, Hardikar, Gonzalez Alam, Bernhardt, Wang, Strawson, Milham, Xu, Margulies, Poerio, Jefferies, Skipper et al., 2025), to reporting across days and months on wide range of experiences (Wang, Bas, Janet, Smallwood, Xu, Morawetz and Tusche, 2026a). For naturalistic paradigms, this typically involves collecting subjective dimensional ratings of the media. These ratings can be unidimensional (e.g., sustained attention, aesthetic appeal) (Song et al., 2021a; Isik and Vessel, 2019) or multidimensional (e.g., arousal and valence) (Ke et al., 2025).

Figure 2a illustrates a classic dual-axis continuous rating system for emotion, situated within a circular space defined by two dimensions (of course, this can be applied to any psychological variables, not just emotional rating). This follows the circumplex model of affect (Gurtman, 1994), which describes emotional states using valence (ranging from unpleasant to pleasant) and arousal (ranging from low to high energy) (Russell, 2003). In this setup, ratings are continuous and participants are instructed to provide real-time, relative ratings of their feelings continuously while viewing the stimulus.

Subsequently, various post-hoc analytical methods can be applied to the collected ratings (Figure 2b). A common approach involves calculating the grand average of group ratings to derive a "consensus" score at the population level; in the context of emotion, such a score could represent the arousal and valence evoked by the stimulus perceived at population level. Additionally, researchers can compute Intersubject Correlation (ISC) between ratings to assess the temporal consistency of emotional experiences across participants (Ke et al., 2025; Lahnakoski, Glerean, Salmi, Jääskeläinen, Sams, Hari and Nummenmaa, 2012; Jung, Amini, Hunt, Murphy, Sadil, Halchenko, Petre, Miao,

Kragel, Han, Heilicher, Sun, Collins, Lindquist and Wager, 2025; Morgenroth, Moia, Vilaclara, Fournier, Muszynski, Ploumetsakou, Almató-Bellavista, Vuilleumier and Van De Ville, 2025). Ultimately, these continuous behavioral trajectories can guide the analysis of neural signals, for instance, by serving as behavioral measure for dynamic network analysis, connectome-based predictive modeling (CPM), or other models tailored to the researcher's specific objectives (Song, Park, Park and Shim, 2021b; Shen, Finn, Scheinost, Rosenberg, Chun, Papademetris and Constable, 2017). In ANDrate toolbox, we have implemented both ISC and averaged group rating methods.

When collecting continuous behavioral ratings, it is crucial to define the rating dimensions and scales based on the specific research goals (e.g., unidimensional vs. multidimensional scales). A general guideline is to consider the nature of the psychological construct being measured. If the construct spans a latent space defined by the interaction of two dimensions, and if a two-dimensional scale provides a richer description than either dimension alone, a 2D interface should be utilized. Conversely, if the construct is linear, a 1D scale is preferable. Furthermore, the sampling frequency must be determined. A practical rule of thumb is to balance data redundancy with computational efficiency. For example, in fMRI studies, collecting behavioral data at an extremely high frequency may be unnecessary given the inherent constraints of the hemodynamic response's temporal resolution (Logothetis, 2008). In all, these parameters should be calibrated to the specific aims of the study.

2.2. Event boundary detection

Continuous rating reveal the unfolding process of individual experience; however, they do not fully capture how the brain understands and constructs an integrated narrative representation. Therefore, event boundaries can be a useful complementary metric for examining behavior. Research on event boundaries traces back to the pioneering work on event cognition and Event Segmentation Theory by Jeff Zacks and colleagues (Zacks and Tversky, 2001; Zacks and Swallow, 2007; Kurby and Zacks, 2008). Simply put, an event boundary is defined as a spontaneous cognitive rupture point occurring whenever a subjective "event" transition is perceived, such as a major change in topic, location, or time. From a cognitive perspective, these boundaries represent critical points where the brain parses a continuous stream of experience into discrete, meaningful segments. Triggered by prediction errors resulting from sudden situational changes, these boundaries prompt the brain to update its active mental models and "chunk" information into long-term memory (Sargent, Zacks, Hambrick, Zacks, Kurby, Bailey, Eisenberg and Beck, 2013; Silva, Wu, Sabio, Conde-Blanco, Roldán, Donaire, Carreño, Axmacher, Baldassano and Fuentesmilla, 2025; DuBrow and Davachi, 2013).

Boundary segmentation has been employed in numerous psychological studies, including research on episodic memory in time compression (Jeunehomme and D'Argembeau, 2020), the impact of domain knowledge on segmentation and memory (Feller, Kurby, Newberry, Schwan and Magliano, 2023), the neural basis of continuous event segmentation in narratives (Baldassano et al., 2017), and how event representations are related to schemas (Baldassano, Hasson and Norman, 2018). These studies usually involve asking annotator to identify timestamps of major narrative transitions during naturalistic stimuli, and calculate consensus boundaries at group level across multiple participants. Figure 3a illustrates a classic paradigm. In a typical segmentation task, participants are instructed to press a button whenever they perceive the end of one natural and meaningful unit of activity and the beginning of another. Additionally, some experiments would also involve naming the events that are formed between boundaries to qualitatively understand event categories and content (Nastase, Liu, Hillman, Zadbood, Hasenfratz, Keshavarzian, Chen, Honey, Yeshurun, Regev, Nguyen, Chang, Baldassano, Lositsky, Simony, Chow, Leong, Brooks, Micciche, Choe and Hasson, 2021; Lee and Chen, 2022; Heusser, Fitzpatrick and Manning, 2021). Despite being given minimal guidance and told to rely on their own judgment, participants often press buttons at similar time points, with high agreement on event descriptions, leading to moments that the majority of participants recognize as event boundaries (Zacks and Tversky, 2001; Zacks and Swallow, 2007). Therefore, it is possible to infer of how the mind process and understand continuous stream of experience. These common experience were then inferred through clustering overlapping boundaries to derive consensus group boundaries (Figure 3b).

Subsequently, various post-hoc analytical methods can be applied to the collected event boundary timestamps. For example, clustering individual boundary to discover consensus boundary. In ANDrate, we implemented the traditional approach for computing consensus event boundaries (Figure 3b). Simply speaking, this method involves setting a pre-defined threshold to evaluate the alignment of boundary timestamps across participants; a consensus boundary is identified if the density of timestamps at a specific moment exceeds the threshold. Following this, the semantic meaning of the event at that boundary can be inferred from the collective textual labels provided by the cohort.

It's worth to mention that, alternative computational methods exist, such as peak-detection-based approaches (Sasmitha and Swallow, 2023) and Hidden Markov Model (HMM) clustering for event discovery (Heusser et al.,

Table 1
Comparison of Common Behavioral Metrics in Naturalistic Experiments

Feature	Continuous Rating	Event Boundary Detection
Core Behavioral Measure	Dynamic tracking of subjective experiences (e.g., valence, arousal).	Identification of discrete transition points (segmentation) in a narrative.
Theoretical Basis	Continuous Behavior rating; Experience Sampling.	Event Segmentation Theory (EST); Mental model updating.
Data Format	Time series.	Event timestamps (onset/offset). Text Labels
Consensus Algorithms	Group Average; Intersubject Correlation (ISC).	Timestamps overlap.

2021; Raccach, Chen, Gureckis, Poeppel and Vo, 2024). However, given the vast possibilities of post-hoc analysis, it's impossible to implement all methods and falls outside the scope of the ANDrate toolbox. So, we have only incorporated the most fundamental consensus boundary detection method, leaving other advanced analytical techniques to the user's decision.

3. Collecting behavioral data in ANDrate

ANDrate is written in python, and is provided in GUI format for researchers. Therefore, collecting behavioral data in ANDrate is highly intuitive. The following sections will walk through a step-by-step guide to ANDrate toolbox, including configuration, data collection, post-rating analysis and visualization. For those who wish to modify for their own use, we also include an overview of the corresponding code implementations at the end of each section.

ANDrate is a lightweight web application based on Flask. The main entry point of the project is `app.py`, which is responsible for launching the local server and providing functionalities such as experiment configuration, eye-tracking calibration, behavioral segmentation, continuous rating, data saving, and subsequent analysis. The frontend pages are primarily stored in the `templates/` folder, while shared frontend scripts are stored in the `static/` folder. The overall structure of the project is as follows:

```

ToolBox/
  app.py           # Flask main program, handles server and API routes
  requirements.txt  # Python dependencies
  webgazer.js      # Local WebGazer eye-tracking library
  precision_calculation.js # Auxiliary JavaScript file

templates/
  index.html       # Experiment configuration page
  calibration.html  # Eye-tracking calibration page
  segmentation.html # Behavioral segmentation task page
  rating.html      # Continuous rating task page
  visualization.html # Post-rating data analysis page

static/
  media.js         # Shared media playback logic
  calibration_core.js # Shared eye-tracking calibration logic

uploads/          # Stores uploaded video or audio stimuli
data/             # Stores collected behavioral and eye-tracking data
mediapipe/        # WebGazer/MediaPipe related model resources
misc/             # Other auxiliary files

```

Among these, `uploads/` is used to store the video or audio stimuli played during the experiment, and `data/` serves as the root directory for behavioral data, where .CSV files generated for each participant are saved. When collecting

data in practice, ANDrate can save files directly under `data/`, or organize them into corresponding subfolders based on experimental materials or sessions, such as `data/dme/`.

3.1. Step 1: Base Parameter Configuration

ANDrate requires configuring several parameters before starting collection, which are divided into two main panels within the Configuration interface (Figure 4), namely `Meta Information` and `Global Task Settings`. Within the `Meta Information` panel, the configuration primarily defines the data storage formats and file naming conventions. This panel comprises three key options where the `Subject Prefix` and `Subject ID` jointly determine the output filename. For instance, given a prefix of `"dme_"` and a subject ID of `01`, the generated file will be saved as either `dme_Sub01_rating.csv` or `dme_Sub01_seg.csv`, depending on the specific task type. Furthermore, the `Save File Name` parameter specifies the root directory. If this is designated as `"Video1"`, the resulting data structure will follow a specific hierarchy where the `"Video1"` root folder contains all corresponding subject subdirectories. In addition to filename configuration, users must select the media files for playback within this initial setup phase. Both video and audio stimuli are supported. Media files can be uploaded via two equivalent methods, the first of which is direct uploading through the web interface as displayed under `Upload New Video`. Alternatively, files can be manually placed into the local `'uploads'` directory within the toolbox. These two methods are functionally identical, as files uploaded via the web interface are automatically routed to the local `'uploads'` folder for storage and access.

In the `Global Task Settings` panel, users define the continuous rating dimensions, sampling frequency, and the activation status of the eye-tracking module based on their experimental design. Here, eye tracking is based on a web camera, supported by `WebGazer.js` (Papoutsaki, Sangkloy, Laskey, Daskalova, Huang and Hays, 2016), and primarily serves as a tool for attention detection. As a cost-effective alternative to dedicated hardware, webcam-based eye tracking has demonstrated robust capabilities in monitoring user attention while maintaining low financial overhead. Prior research indicates that under well-controlled environmental conditions, webcam tracking can achieve comparable efficacy to professional eye trackers (Kim, He, Wu, Dai, Kohlhoff, Turner, Paul, Kennedy, Adolphs and Navalpakkam, 2024; Levordashka, Richardson, Hirst et al., 2025; Hutt, Wong, Papoutsaki et al., 2024). However, it is important to note that webcam solutions remain susceptible to substantial ambient noise, rendering them suitable primarily for eye-tracking studies with coarse-grained precision requirements. Following the recommendations of Van der Cruyssen et al. (Van der Cruyssen, Ben-Shakhar, Pertzov et al., 2024), we integrated webcam eye tracking into ANDrate primarily as a data-quality verification mechanism (e.g., establishing Areas of Interest to verify whether participants remain focused on the task). Further implementation details are discussed in the post-rating analysis.

For `Event Boundary Segmentation` tasks, users must configure the `Segmentation Sampling Frequency` parameter in milliseconds, which establishes the temporal resolution for user-generated timestamps. For example, a sampling frequency of 1000 ms restricts the event boundary timestamps to 1-second increments. For `Continuous Rating` tasks, users also need to configure the `Sampling Frequency` parameter in milliseconds, which establishes the temporal resolution for collecting continuous ratings. Besides, the required parameters also include the names and ranges for both the `X-axis` and `Y-axis`. In cases where a one-dimensional rating task is conducted, users can select the `Disable Y Axis Rating` checkbox. Finally, if the `Eye Tracking Module` is enabled, a standard 9-point calibration procedure is automatically initiated prior to each behavioral rating session. Consequently, researchers must specify the `Calibration Accuracy Threshold`, which defines the minimum accuracy required for a participant to proceed. If the calibration accuracy falls below this threshold, the participant is prompted to recalibrate. In cases where the calibration fails consecutively three times, the procedure terminates automatically.

The workflow pseudocode for this module is presented below. The code on configuration can be found under `templates/index.html`. The eye tracking calibration can be found under `templates/calibration.html`, `static/calibration_core.js` and `precision_calculation.js`.

Configuration and Data Storage

```

1: Step 1: Configure file naming
2: filename ← subjectPrefix+“Sub”+subjectId+“_”+taskType+“.csv”
3: outputPath ← saveFolder+“/”+filename
4: Step 2: Upload media files
5: Support video: .mp4/.avi/.mov/.mkv/.webm
6: Support audio: .mp3/.wav/.m4a/.aac/.ogg/.flac/.wma/.opus
7: Media can be uploaded via webpage or manually placed in uploads/
8: Step 3: Configure global settings
9: Set rating dimension, sampling frequency, and eye-tracking option
10: Step 4: Configure task-specific settings
11: if taskType = Segmentation then
12:   Set Segmentation Sampling Frequency
13: else if taskType = Continuous Rating then
14:   Set Sampling Frequency, X/Y axis labels and ranges
15:   if 1D rating enabled then
16:     Disable Y-axis rating
17:   end if
18: end if
19: Step 5: Configure eye tracking
20: if eye tracking enabled then
21:   Start 9-point calibration
22:   Set Calibration Accuracy Threshold
23:   if 3 consecutive calibration failures then
24:     Terminate calibration
25:   end if
26: end if
27: Step 6: Save data
28: POST /api/save_csv
29: Payload: { filename, csvContent, saveFolder }
30: targetDir ← data+“/”+saveFolder
31: WRITE(targetDir+“/”+filename, csvContent)

```

3.2. Step 2: Event Segmentation and Continuous Rating Data Collection

Prior to each experimental session, there will be an eye-tracking calibration procedure if that the eye-tracking module is enabled within the configuration settings. We adopts the standard nine-point calibration protocol recommended by WebGazer.js (Papoutsaki et al., 2016). During this procedure, participants are instructed to sequentially click on nine distinct calibration anchors distributed across the screen while maintaining visual fixation on the cursor. Upon completion of the calibration, participants are directed to fixate on a centralized validation point on the screen, during which their gaze coordinates are sampled across 50 consecutive frames. The tracking accuracy is quantified by calculating the ratio of the average geometric distance (between the predicted gaze points and the center validation point) to half of the screen height. The final accuracy score is presented as a rounded percentage; a higher score denotes higher calibration precision. Once the eye-tracking accuracy meets the predefined threshold, the system transitions to the formal behavioral data collection phase. For the event segmentation task, the user interface is depicted in Figure 5. The interface contains four functional zones: the Media Player Zone, the Timeline/Progress Bar Zone, the Function Bar Zone, and the Eye-Tracking Monitoring Zone.

The Media Player Zone serves as the primary presentation device, where participants attend to the audiovisual stimuli. During the event segmentation task, participants can perform real-time boundary marking by pressing the designated “b” key on the keyboard. The temporal resolution of the logged timestamps is governed by the configuration parameters established in Step 1. When a boundary is registered, it would populate the Boundary Table within the Function Bar Zone on the right. Consequently, an “event” is created between two consecutive boundaries, which is reflected by the corresponding temporal range in the boundary table. Within this workspace, participants are permitted

to delete specific boundaries or assign descriptive text labels to them. To facilitate labeling, left-clicking an event automatically repositions the playback to its start time. In practice, this feature is expected to assist participants in naming the boundaries more efficiently.

Lastly, the Eye-Tracking Monitoring Zone provides real-time feedback regarding the user's fixation. This zone consists of a monitoring window integrated with the webcam feed alongside corresponding control buttons. Crucially, the border color of this window dynamically shifts based on the specific screen region the user is currently fixating on (e.g., a green border for the video area, a blue border for the function bar, an orange border for the timeline, and yellow border for all other regions). Although real-time visualizations of both the webcam feed and the gaze point cursor are enabled by default, users retain the autonomy to manually toggle them off. Notably, the underlying gaze data continues to be recorded in the background without interruption.

Upon completing the event segmentation task, participants can export the data, which would download the behavioral data locally within the /data directory according to the metadata schema defined in Step 1. By default, segmentation data are saved with the suffix `_seg.csv`, whereas concurrent eye-tracking data, if recorded, are appended as `_seg_gaze.csv`. The segmentation data file are detailed in Figure 6. Specifically, the segmentation output logs the onset and offset timestamps of each discrete event alongside its corresponding text label. The eye tracking data file contains the timestamp of each eye-tracking sampling frame, the X and Y spatial coordinates of the gaze point, the AOI the gaze point belongs to (`On_Video`, `On_Function`, `On_Time`, or `Other`), and the specific calibration accuracy score associated with that session (Figure 7).

Similar to the interface for the segmentation task, there are four functional zones in the continuous rating task (see Figure 8). The only difference is that the Function Bar Zone is now substituted by a two-dimensional coordinate axis designed for continuous affective evaluation (only one x-axis if y-axis is disabled). The specific dimensions of this rating scale are dictated by the initial configuration profiles. At the baseline, the rating cursor is initialized at the origin (0, 0). Participants can use either the keyboard directional arrow keys or directly use the mouse cursor to manipulate the rating cursor, thereby providing continuous ratings alongside the media. This continuous trajectory is visualized dynamically beneath the timeline area. Upon completing the continuous rating task, the data would be saved as `_rating.csv` file, and a `_rating_gaze.csv` file if eye-tracking is enabled. The rating data file contains the time units (in ms) determined by the sampling frequency and the corresponding rating scores, where the variable names correspond exactly to the configured names of the X and Y axes (Figure 9). The eye tracking data format is similar to the event segmentation task (Figure 7).

The code for segmentation can be found under `templates/segmentation.html`, and the code for continuous rating can be found under `templates/rating.html`. The core workflow of the data collection module is presented below:

Behavioral and Eye-Tracking Data Collection

```

1: Load task configuration and media file
2: if eye tracking enabled then
3:   Initialize WebGazer tracker
4:   Perform 9-point calibration
5:   Validate calibration accuracy
6:   if accuracy below threshold then
7:     Recalibrate or terminate
8:   end if
9: end if
10: Start media playback
11: if task = Event Segmentation then
12:   Record boundary timestamps and labels
13: else if task = Continuous Rating then
14:   Record continuous rating values
15: end if
16: if eye tracking enabled then
17:   Record gaze timestamp and gaze coordinates
18:   Classify gaze region
19: end if
20: Save and POST behavioral and gaze data to /api/save_csv

```

3.3. Step 3: Post-Data Collection Analysis and Visualization

We implemented a post-rating data analysis and visualization pipeline within the ANDrate toolbox. As illustrated in Figure 10, the toolbox supports five distinct analytical and visualization workflows. The users are simply required to import the collected datasets (with eye-tracking files being optional) and specify the corresponding stimulus material presented during the experiment. Data import can be executed in two ways. Either automatic, wherein the toolbox recursively parses the root directory (/data) to identify matching folder architectures, or manual, which requires the explicit selection of target segmentation and/or rating files. Users can subsequently choose from the preferred analytical pipeline, which is as follows:

First, for the eye-tracking data, the system will calculate the percentage of fixations allocated to three predefined areas of interest (AOIs) on the data collection interface: the Media Player Zone, the Timeline/Progress Bar Zone, and the Function Bar Zone (the Eye-Tracking Monitoring Zone is included here). Any fixation falling outside these designated AOIs is categorized as 'Other'. Then, a distribution map of attention allocation and calibration accuracy scores for each subject is presented. Given that fixations generally reflect attentional engagement, they serve as an effective instrument for attention checks. Furthermore, due to the inherent noise of webcams, an AOI-based approach is the most appropriate analytical method (Van der Cruyssen et al., 2024). We expect these metrics to constitute an effective tool for data quality control.

Second, for the continuous rating analysis, the toolbox provides two analytical approaches: the Inter-Subject Correlation (ISC) module and the Group Average module. The ISC module quantifies the synchronization of dimensional ratings across participants by computing the Pearson correlation coefficient between each leave-one-out individual's time-series rating and the average of the remaining cohort, thereby allowing researchers to assess the alignment of participants' emotional or cognitive experiences. Concurrently, the Group Average module computes the arithmetic mean of the cohort's ratings over time and superimposes this aggregate trajectory directly onto the video timeline to pinpoint precisely when specific group-level experience shifts occur during naturalistic stimulation. For the visualization, the moment-to-moment group-averaged rating is presented alongside the media. In addition, the toolbox also displays each leave-one-out participant's correlation coefficient as well as the averaged ISC score using a histogram. We expect these metrics to facilitate the moment-to-moment analysis of experiential dynamics.

Third, regarding the event segmentation synthesis, the toolbox implements an algorithm based on the inter-subject overlap rate to identify consensus boundaries. Enabling this analysis introduces two additional configuration parameters on the user interface. The first is the "Consensus boundary window," which defines the temporal window size for counting overlapping boundaries; for instance, a setting of ± 4 seconds dictates that timestamps falling within this

range are considered coincident. The second is the "Min rater overlap," which specifies the minimum percentage of participants required to share a boundary for it to qualify as a consensus. For example, a threshold of 50 implies that a cluster must encompass at least half of the total participants. Concurrently, participants' textual event labels are aggregated and visualized as semantic word clouds for each consensus event, thereby facilitating the interpretation of their underlying content. Within the visualization interface, both individual-level and consensus boundaries, along with their respective consensus and individual labels, are explicitly displayed. We expect this metric to facilitate analysis on how the human mind parses continuous information streams and understand it in units of events.

Fourth, the toolbox offers a summary analysis that overlays both consensus boundaries and consensus continuous rating trajectories, providing an intuitive and direct representation of the relationship between discrete event segmentation and continuous experience sampling. Specifically, this feature visualizes the event boundaries as vertical black dashed lines overlaying the continuous rating trajectories, with the boundary parameter configurations remaining consistent with those described above. We expect that this integrated visualization will provide researchers with a comprehensive behavioral profile for the naturalistic stimuli.

The core computational workflow of the post-collection analysis module is summarized in the following pseudocode. The code can be found under templates/visualization.html. This workflow integrates automated data loading, AOI-based gaze quality control, inter-subject synchronization analysis, consensus boundary detection, and multimodal visualization into a unified analytical pipeline, thereby enabling quantitative investigation of attentional allocation, shared experiential dynamics, and event-structure agreement across participants.

Post-collection Analysis and Visualization

```

1: Load segmentation files, rating files, optional gaze files, and stimulus media
2: if mode = auto then
3:   Recursively scan /data and match files by suffix
4: else
5:   Load user-selected files
6: end if
7: Parse segmentation, rating, and gaze datasets by participant
8: if gaze data available then
9:   for each gaze sample do
10:    Classify region as On_Video, On_Time, On_Function, or Other
11:   end for
12:    $AOI\% \leftarrow n_{AOI}/n_{total} \times 100$ 
13:   Store calibration accuracy for each participant
14: end if
15: Align rating time series according to sampling timestamps
16:  $GroupMean(t) \leftarrow \frac{1}{N} \sum_{i=1}^N Rating_i(t)$ 
17: for each participant  $i$  do
18:    $OthersMean_i(t) \leftarrow \frac{1}{N-1} \sum_{j \neq i} Rating_j(t)$ 
19:    $ISC_i \leftarrow corr(Rating_i(t), OthersMean_i(t))$ 
20: end for
21:  $MeanISC \leftarrow \frac{1}{N} \sum_{i=1}^N ISC_i$ 
22: Cluster boundary timestamps within  $\pm window$  seconds
23: for each boundary cluster  $c$  do
24:    $Overlap(c) \leftarrow n_{raters}(c)/N \times 100$ 
25:   if  $Overlap(c) \geq minOverlap$  then
26:     Retain  $c$  as consensus boundary
27:   end if
28: end for
29: Collect labels around each consensus boundary
30: Generate semantic word clouds from aggregated labels
31: Overlay consensus boundaries on group-mean rating trajectory
32: Export AOI report, calibration scores, ISC results, group ratings, consensus boundaries, and summary figures

```

4. Practical Exercises

To better demonstrate the utility of the present toolbox, we conducted an exemplary experiment covering the entire workflow as practical exercises. This includes experimental data collection, results analysis, visualization, and finally discussion.

Visual stimuli were adapted from the NATVIEW dataset (Telesford, Gonzalez-Moreira, Xu, Tian, Colcombe, Cloud, Russ, Falchier, Nentwich, Madsen, Parra, Schroeder, Milham and Franco, 2023), which included a clip from the animated movie *Despicable Me* (a 10-minute edited version), a short film titled *The Present*, and a documentary video featuring macaques (referred to hereafter as the "Monkey video"). These stimuli were selected due to their distinct emotion profile: *Despicable Me* are rich in dramatic conflicts, so it has highly intense and dynamic emotional content, whereas in *The Present*, the emotional intensity was generally weaker. Conversely, the Monkey video served as a baseline control with minimal emotional fluctuations. Based on these characteristics, we hypothesized that the fluctuations in both valence and arousal would be most pronounced in *Despicable Me*. In contrast, *The Present* was expected to exhibit marked fluctuations primarily in valence rather than arousal, while the Monkey video would display relatively stable trajectories across both dimensions. Regarding event segmentation, we further hypothesized that event boundaries would consistently demarcate transitions in affective experiences across the stimuli. While these stimuli were video-based, the toolbox is equally applicable to auditory materials. Detailed descriptions and access to these video materials can be found within the repository of the NATVIEW dataset.

A total of 14 raters participated in the validation experiment. 10 participants contain eye-tracking data. Regarding the parameter configurations within the toolbox, the segmentation sampling frequency was set to 1000 ms, and the rating sampling frequency was set to 200 ms, and participants were instructed to perform continuous two-dimensional affective ratings across the Valence and Arousal dimensions. The webcam-based eye-tracking module was enabled with a Calibration Accuracy threshold set to 60. The Min rater overlap was set to 50 % of raters in a cluster with a window size of ± 6 in the post-analysis setting.

Each participant viewed the three video clips twice to complete two behavior tasks: event segmentation and continuous affective rating. In the segmentation task, participants were instructed to segment the movies based on their own perceived event boundaries, without any explicit constraints (e.g., no instructions forcing coarse- or fine-grained segmentation). In the rating task, the participants were instructed to perform continuous ratings in two affect dimensions, arousal and valence. The underlying psychological meanings of these affective dimensions were explained beforehand, and participants were instructed to report their real-time emotional experiences while watching the movies.

Results and Discussion

The experimental results are illustrated in Figure 11. First, eye-tracking analysis revealed that among the 10 participants with eye-tracking data, the majority achieved a calibration accuracy above 70%. Specifically, the group average calibration accuracy reached 73.2% for *Despicable Me*, 73.3% for *The Present*, and 73.8% for the Monkey video. In terms of gaze distribution, participants spent most of their time looking directly at the video area, with group averages of 58.0% for *Despicable Me*, 58.8% for *The Present*, and 58.2% for the Monkey video. This confirms that the majority of participants complied with the experimental instructions and remained actively engaged with the video stimuli during the tasks.

However, the analysis also revealed substantial individual differences in attentional allocation across different videos. For instance, Participant 11 failed to focus on the video content during *Despicable Me*, spending only 34.7% of the time on the video area while dwelling on the time bar for 37.3% of the trial. This gaze pattern may suggest a lack of task compliance. Based on this observation, we established an exclusion criterion for subsequent behavioral analyses: data from participants whose "On Video" dwell time was below 50% or failed to constitute the highest proportion of gaze allocation were excluded. Consequently, three participants were excluded for *Despicable Me* (Participants 5, 11, and 12), three for *The Present* (Participants 8, 13, and 14), and three for the Monkey video (Participants 7, 8, and 14). Detailed gaze metrics are available in the Supplementary Figures S1 to S3.

Notably, no further analyses were performed on the remaining data. As mentioned above, although WebGazer.js is widely adopted in online research, it relies on a standard webcam rather than professional high-speed eye-trackers, inherently limiting its spatial and temporal precision. In practice, the vast majority of studies utilizing WebGazer.js restrict their scope to Area of Interest (AOI) statistics, which aligns precisely with the analytical approach implemented in this study.

Regarding the continuous rating data, we found that at the group level, consensus affective experiences closely tracked the narrative progression of the videos as expected. Figure 12a suggested that, in *Despicable Me*, Valence and Arousal scores plummeted to a nadir when the protagonist, Gru, was forced to send the girls away, and subsequently peaked when Gru took the rocket for his moon-stealing plan. Similarly, in *The Present*, a sharp drop in ratings occurred when the boy discovered that the puppy gifted to him had only one leg; however, the valence ratings gradually recovered as he began playing with the puppy (Figure 12b). Intriguingly, compared to *Despicable Me*, *The Present* lacks intense dramatic conflicts, which is reflected in the absence of drastic fluctuations in the arousal rating curves. Validating this even further, the Arousal ratings for the Monkey video remained close to zero, as the stimulus comprised merely a compilation of interactions between Monkeys (see Supplementary Figure S4 for details).

Furthermore, Inter-Subject Correlation (ISC) analysis demonstrated that participants achieved the highest synchrony in both Valence and Arousal dimensions during *Despicable Me* (mean $r_{\text{Valence}} = 0.698$; mean $r_{\text{Arousal}} = 0.716$). For *The Present*, highly synchronized patterns were preserved in the Valence dimension (mean $r_{\text{Valence}} = 0.702$), whereas the synchrony for Arousal substantially declined (mean $r_{\text{Arousal}} = 0.269$). Validating this further, the valence and arousal synchrony dropped further (see Supplementary Figure S4 for details). Collectively, these findings highly align with our hypotheses. By manipulating the narrative properties of the video stimuli, we demonstrated that ANDrate can successfully capture continuous, fine-grained affective tracks across individuals and effectively isolate consensus emotional experiences.

For the event segmentation analysis, the results also indicate that ANDrate is highly capable of identifying meaningful consensus event boundaries. As illustrated in Figure 13, for *Despicable Me*, the toolbox successfully identified event corresponding to the scene where Gru reads a bedtime story to the girls, alongside with keywords in the word cloud (e.g., “Story” “Telling” and “bedtime”). Similarly, for *The Present*, it identified event where the boy recognizes the shared disability between himself and the puppy and decides to play outside (“the boy playing” “puppy”). For the Monkey video, it captured the event about a man buying monkeys from a merchant (“Monkeys,” “transaction” and “trading”).

Importantly, these boundaries also marks a transition in affective experience. A joint analysis revealed that emotional ratings exhibited pronounced changes around these identified consensus boundaries. For example as shown in Figure 14, in *Despicable Me*, boundaries like Minions making troubles and the moon-stealing begin was accompanied by a concurrent surge in both Valence and Arousal scores (see Supplementary Figures S5 and S6 for corresponding visualizations of the other videos). This phenomenon strongly aligns with Event Segmentation Theory. Specifically, because an event boundary serves as a cognitive transition point where the brain parses continuous ecological experiences into discrete episodes, shifts in emotional valence or intensity frequently constitute the catalyst for these cognitive boundaries (Chen and Swallow, 2025; McClay, Sachs and Clewett, 2023).

Taken together, the results demonstrate that ANDrate is an effective tool for naturalistic stimuli, which integrates behavioral data collection and post-hoc rating analysis within a single ecosystem. Crucially, all the aforementioned analyses can be executed directly within the Graphical User Interface (GUI), allowing users to export processed data (such as consensus boundaries and time-series ratings) for subsequent neuroimaging analyses or alternative computational workflows. In conclusion, ANDrate represents a highly flexible, user-friendly, and powerful tool tailored for researchers in psychology and cognitive neuroscience.

5. Limitation and Future Work

The present study is subject to several limitations that also point toward directions for future research. First, while this tutorial demonstrates the local deployment of ANDrate, it has not yet been implemented as a cloud-based solution, making it difficult to leverage online crowdsourcing platforms to scale up the data collection process. In fact, we attempted to deploy the toolbox on a behavioral data collection server developed within our laboratory. However, due to various factors, including video latency issues and ethical and data security considerations involved in hosting other researchers’ data on our platform, the cloud-based version has not yet been released for public use. Nonetheless, we have made the source code of ANDrate open-source, allowing researchers to perform their own cloud deployments. Future development will focus on integrating the toolbox with online experimental frameworks to facilitate accelerated data collection via crowd sourcing platforms.

6. Conclusion

Naturalistic analysis plays an increasingly vital role in psychology and cognitive neuroscience, enabling the study of psychological phenomena in ways that mirror real-world human behavior. While these analyses have been extensively applied at both behavioral and neural levels, the naturalistic paradigm has often lacked explicit behavioral measures and therefore has to rely on external raters. In this tutorial, we have provided an intuitive and comprehensive overview of methods for collecting behavioral measures for naturalistic paradigms and developed ANDrate, a user-friendly toolbox designed for this purpose. Furthermore, we have provided a step-by-step guide covering the entire workflow for practical applications, including typical experimental design, data collection, and post-rating analysis and visualization. To facilitate the adoption of the toolbox, we utilized ANDrate to collect and analyze behavioral data, providing accessible examples and making the associated datasets and scripts publicly available. We believe that researchers in the fields of psychology and cognitive neuroscience will find this tutorial and the accompanying toolbox to be valuable assets for their research.

Declarations

Funding

This work was mainly supported by the Science and Technology Development Fund (FDCT) of Macau [0112/2024/RIA2], and the MYRG of University of Macau (MYRG-GRG2025–00175-ICI).

Conflicts of interest

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics Approval

The experimental protocol of this study was approved by the ethical review committee of University of Macau (BSERE21-APP006-ICI).

Consent to Participate

Informed consent was obtained from all individual participants included in the study.

Consent for publication

The manuscript is approved by all authors for publication.

Availability of data and materials

ANDrate is open-sourced under the GNU General Public License (GPL) and can be downloaded for free from the lab GitHub at <https://github.com/andlab-um/ANDrate>. The data for reproducing this paper is also contain in the link.

Author Contributions

E.R.W.W and H. Wu conceived the research, E.R.W.W and J. Rui developed the toolbox, E.R.W.W performed the research, E.R.W.W analyzed the data, E.R.W.W and J. Rui, and H. Wu wrote the paper.

References

- Baldassano, C., Chen, J., Zadbood, A., Pillow, J.W., Hasson, U., Norman, K.A., 2017. Discovering event structure in continuous narrative perception and memory. *Neuron* 95, 709–721.e5.
- Baldassano, C., Hasson, U., Norman, K.A., 2018. Representation of real-world event schemas during narrative perception. *The Journal of Neuroscience* 38, 9689–9699.
- Chen, J., Leong, Y.C.M., Honey, C.J., Yong, C.H., Norman, K.A., Hasson, U., 2017. Shared memories reveal shared structure in neural activity across individuals. *Nature Neuroscience* 20, 115–125.
- Chen, R., Swallow, K.M., 2025. The role of emotional content in segmenting naturalistic videos into events. *Journal of Experimental Psychology: General* 154, 2158–2178. doi:10.1037/xge0001783.
- Christensen, T.C., Barrett, L.F., Bliss-Moreau, E., et al., 2003. A practical guide to experience-sampling procedures. *Journal of Happiness Studies* 4, 53–78. URL: <https://doi.org/10.1023/A:1023609306024>.
- Conner, T.S., Tennen, H., Fleeson, W., Barrett, L.F., 2009. Experience sampling methods: A modern idiographic approach to personality research. *Soc Personal Psychol Compass* 3, 292–313. doi:10.1111/j.1751-9004.2009.00170.x. PMID: 19898679; PMCID: PMC2773515.

- Van der Cruyssen, I., Ben-Shakhar, G., Pertzov, Y., et al., 2024. The validation of online webcam-based eye-tracking: The replication of the cascade effect, the novelty preference, and the visual world paradigm. *Behavior Research Methods* 56, 4836–4849. doi:10.3758/s13428-023-02221-2.
- DuBrow, S., Davachi, L., 2013. The influence of context boundaries on memory for the sequential order of events. *Journal of Experimental Psychology: General* 142, 1277–1286.
- Feller, D.P., Kurby, C.A., Newberry, K.M., Schwan, S., Magliano, J.P., 2023. The effects of domain knowledge and event structure on event processing. *Memory & Cognition* 51, 101–114.
- Girard, J.M., 2014. Carma: Software for continuous affect rating and media annotation. *Journal of Open Research Software* 2, e5.
- Girard, J.M., Wright, A.G.C., 2018. Darma: Software for dual axis rating and media annotation. *Behavior Research Methods* 50, 902–909.
- Gurtman, M.B., 1994. The circumplex as a tool for studying normal and abnormal personality: A methodological primer. *Journal of Personality Assessment* 62, 344–361.
- Heusser, A.C., Fitzpatrick, P.C., Manning, J.R., 2021. Geometric models reveal behavioural and neural signatures of transforming experiences into memories. *Nature Human Behaviour* 5, 905–919. doi:10.1038/s41562-021-01051-6.
- Hutt, S., Wong, A., Papoutsaki, A., et al., 2024. Webcam-based eye tracking to detect mind wandering and comprehension errors. *Behavior Research Methods* 56, 1–17. doi:10.3758/s13428-022-02040-x.
- Isik, A.I., Vessel, E.A., 2019. Continuous ratings of movie watching reveal idiosyncratic dynamics of aesthetic enjoyment. *PLoS One* 14, e0223896.
- Jeunehomme, O., D'Argembeau, A., 2020. Event segmentation and the temporal compression of experience in episodic memory. *Psychological Research* 84, 481–490.
- Jung, H., Amini, M., Hunt, B.J., Murphy, E.I., Sadil, P., Halchenko, Y.O., Petre, B., Miao, Z., Kragel, P.A., Han, X., Heilicher, M.O., Sun, M., Collins, O.G., Lindquist, M.A., Wager, T.D., 2025. Spacetop: A multimodal fmri dataset unifying naturalistic processes with a rich array of experimental tasks. *Scientific Data* 12, 1465.
- Ke, J., Song, H., Bai, Z., Rosenberg, M.D., Leong, Y.C., 2025. Dynamic brain connectivity predicts emotional arousal during naturalistic movie-watching. *PLoS Computational Biology* 21, e1012994.
- Kim, N.Y., He, J., Wu, Q., Dai, N., Kohlhoff, K., Turner, J., Paul, L.K., Kennedy, D.P., Adolphs, R., Navalpakkam, V., 2024. Smartphone-based gaze estimation for in-home autism research. *Autism Research* 17, 1140–1148. doi:10.1002/aur.3140.
- Kurby, C.A., Zacks, J.M., 2008. Segmentation in the perception and memory of events. *Trends in Cognitive Sciences* 12, 72–79.
- Lahnakoski, J.M., Glerean, E., Salmi, J., Jääskeläinen, I.P., Sams, M., Hari, R., Nummenmaa, L., 2012. Naturalistic fmri mapping reveals superior temporal sulcus as the hub for the distributed brain network for social perception. *Frontiers in Human Neuroscience* 6, 233.
- Lee, H., Chen, J., 2022. Predicting memory from the network structure of naturalistic events. *Nature Communications* 13, 4235. doi:10.1038/s41467-022-31965-2.
- Levordashka, A., Richardson, M., Hirst, R.J., et al., 2025. Trace: A research media player measuring real-time audience engagement. *Behavior Research Methods* 57, 44. doi:10.3758/s13428-024-02522-0.
- Liu, L., Jiang, J., Li, H., Ding, G., 2025. Tripartite organization of brain state dynamics underlying spoken narrative comprehension. *eLife* 13, RP99997.
- Logothetis, N.K., 2008. What we can do and what we cannot do with fmri. *Nature* 453, 869–878.
- McClay, M., Sachs, M.E., Clewett, D., 2023. Dynamic emotional states shape the episodic structure of memory. *Nature Communications* 14, 6533. doi:10.1038/s41467-023-42241-2.
- Morgenroth, E., Moia, S., Vilaclara, L., Fournier, R., Muszynski, M., Ploumitsakou, M., Almató-Bellavista, M., Vuilleumier, P., Van De Ville, D., 2025. Emo-film: A multimodal dataset for affective neuroscience using naturalistic stimuli. *Scientific Data* 12, 684.
- Nastase, S.A., Goldstein, A., Hasson, U., 2020. Keep it real: rethinking the primacy of experimental control in cognitive neuroscience. *NeuroImage* 222, 117254.
- Nastase, S.A., Liu, Y.F., Hillman, H., Zadbood, A., Hasenfratz, L., Keshavarzian, N., Chen, J., Honey, C.J., Yeshurun, Y., Regev, M., Nguyen, M., Chang, C.H.C., Baldassano, C., Lositsky, O., Simony, E., Chow, M.A., Leong, Y.C., Brooks, P.P., Micciche, E., Choe, G., Hasson, U., 2021. The "narratives" fmri dataset for evaluating models of naturalistic language comprehension. *Scientific Data* 8, 250. doi:10.1038/s41597-021-01033-3.
- Papoutsaki, A., Sangkloy, P., Laskey, J., Daskalova, N., Huang, J., Hays, J., 2016. Webgazer: Scalable webcam eye tracking using user interactions, in: *Proceedings of the 25th International Joint Conference on Artificial Intelligence (IJCAI), AAAI*. pp. 3839–3845.
- Racah, O., Chen, P., Gureckis, T.M., Poeppel, D., Vo, V.A., 2024. The "naturalistic free recall" dataset: four stories, hundreds of participants, and high-fidelity transcriptions. *Scientific Data* 11, 1317. doi:10.1038/s41597-024-04082-6.
- Ruef, A.M., Levenson, R.W., 2007. Continuous measurement of emotion: The affect rating dial, in: Coan, J.A., Allen, J.J.B. (Eds.), *Handbook of emotion elicitation and assessment*. Oxford University Press, New York, NY, US, pp. 286–297.
- Russell, J.A., 2003. Core affect and the psychological construction of emotion. *Psychological Review* 110, 145–172.
- Sargent, J.Q., Zacks, J.M., Hambrick, D.Z., Zacks, R.T., Kurby, C.A., Bailey, H.R., Eisenberg, M.L., Beck, T.M., 2013. Event segmentation ability uniquely predicts event memory. *Cognition* 129, 241–255.
- Sasmita, K., Swallow, K.M., 2023. Measuring event segmentation: An investigation into the stability of event boundary agreement across groups. *Behavior Research Methods* 55, 428–447. doi:10.3758/s13428-022-01832-5.
- Shen, X., Finn, E.S., Scheinost, D., Rosenberg, M.D., Chun, M.M., Papademetris, X., Constable, R.T., 2017. Using connectome-based predictive modeling to predict individual behavior from brain connectivity. *Nature Protocols* 12, 506–518.
- Silva, M., Wu, X., Sabio, M., Conde-Blanco, E.c., Roldán, P., Donaire, A., Carreño, M.C., Axmacher, N., Baldassano, C., Fuentesmilla, L., 2025. Movie-watching evokes ripple-like activity within events and at event boundaries. *Nature Communications* 16, 5647.
- Song, H., Finn, E.S., Rosenberg, M.D., 2021a. Neural signatures of attentional engagement during narratives and its consequences for event memory. *Proceedings of the National Academy of Sciences* 118, e2021905118.

- Song, H., Park, B.y.i., Park, H., Shim, W.M., 2021b. Cognitive and neural state dynamics of narrative comprehension. *Journal of Neuroscience* 41, 8972–8990.
- Telesford, Q.K., Gonzalez-Moreira, E., Xu, T., Tian, Y., Colcombe, S.J., Cloud, J., Russ, B.E., Falchier, A., Nentwich, M., Madsen, J., Parra, L.C., Schroeder, C.E., Milham, M.P., Franco, A.R., 2023. An open-access dataset of naturalistic viewing using simultaneous eeg-fmri. *Scientific Data* 10, 554. doi:10.1038/s41597-023-02458-8.
- Wallace, R.S., Mckeown, B., Goodall-Halliwel, I., Chitiz, L., Forest, P., Karapanagiotidis, T., Mulholland, B., Turnbull, A., Vanderwal, T., Hardikar, S., Gonzalez Alam, T.R.J., Bernhardt, B.C., Wang, H.T., Strawson, W., Milham, M., Xu, T., Margulies, D.S., Poerio, G.L., Jefferies, E., Skipper, J.I., et al., 2025. Mapping patterns of thought onto brain activity during movie-watching. *eLife* 13, RP97731. doi:https://doi.org/10.7554/eLife.97731.
- Wang, E.R.R., Wu, H., 2026. Using movie to understand the human mind: A dual-adaptive framework of natural viewing. *BioRxiv*.
- Wang, R., Bas, L.M., Janet, R., Smallwood, J., Xu, Y., Morawetz, C., Tusche, A., 2026a. Patterns of ongoing thought are associated with emotion regulation and negative affect in daily life. *Affective Science* 7, 1–19. doi:10.1007/s42761-026-00355-y.
- Wang, R., Luo, Y., Wang, Y., Zhang, L., Wu, H., 2026b. Towards naturalistic social neuroscience: A multi-level framework integrating real-world phenotyping, neurobiology, and computational mechanisms. *Cognitive, Affective, & Behavioral Neuroscience*, 1–17doi:10.3758/s13415-026-XXXX-X.
- Wei, Y., Wang, E., 2026. The eye movement pattern and brain dynamics indicate successful learning during education video viewing, in: *Proceedings of the Annual Meeting of the Cognitive Science Society*.
- Xu, X.J., Liu, X., Hu, X., Wu, H., 2025. The trajectory of crime: Integrating mouse-tracking into concealed memory detection. *Behavior Research Methods* 57, 78. doi:10.3758/s13428-024-02594-y.
- Yang, E., Milisav, F., Kopal, J., Holmes, A.J., Mitsis, G.D., Mistic, B., Finn, E.S., Bzdok, D., 2023. The default network dominates neural responses to evolving movie stories. *Nature Communications* 14, 4197.
- Yang, J., Hu, Z., Li, J., Guo, X., Gao, X., Liu, J., Wang, Y., Qu, Z., Li, W., Li, Z., Li, W., Huang, Y., Chen, J., Wen, H., Yuan, B., 2025. Nadyet: A toolbox for dynamic network analysis of naturalistic stimuli. *NeuroImage* 311, 121203. doi:10.1016/j.neuroimage.2025.121203.
- Zacks, J.M., Swallow, K.M., 2007. Event segmentation. *Current Directions in Psychological Science* 16, 80–84.
- Zacks, J.M., Tversky, B., 2001. Event structure in perception and conception. *Psychological Bulletin* 127, 3–21.
- Zhang, S., Seak, L.C.U., Dolan, R.J., Wu, H., 2025. Ai in the computation and regulation of social decision-making. *PsyArXiv* doi:10.31234/osf.io/uf9g4.
- Zhang, Y., Kim, J.H., Brang, D., Liu, Z., 2021. Naturalistic stimuli: A paradigm for multi-scale functional characterization of the human brain. *Current Opinion in Biomedical Engineering* 18, 100272.
- Zheng, J., Chen, X., Wu, H., 2025. An eeg dataset of exploring navigation in virtual reality. *Scientific Data* 12, 1957. doi:10.1038/s41597-025-06175-2.
- Zhou, X., Chen, G., Ning, J., Cai, Y., Zhao, S., Zhao, S., Moulson, M.C., 2026. Facetrack-aoi: An ai-driven tool for automated dynamic aoi placement and eye movement analysis in face perception studies. *Behavior Research Methods* 58, 141. doi:10.3758/s13428-026-XXXX-X.

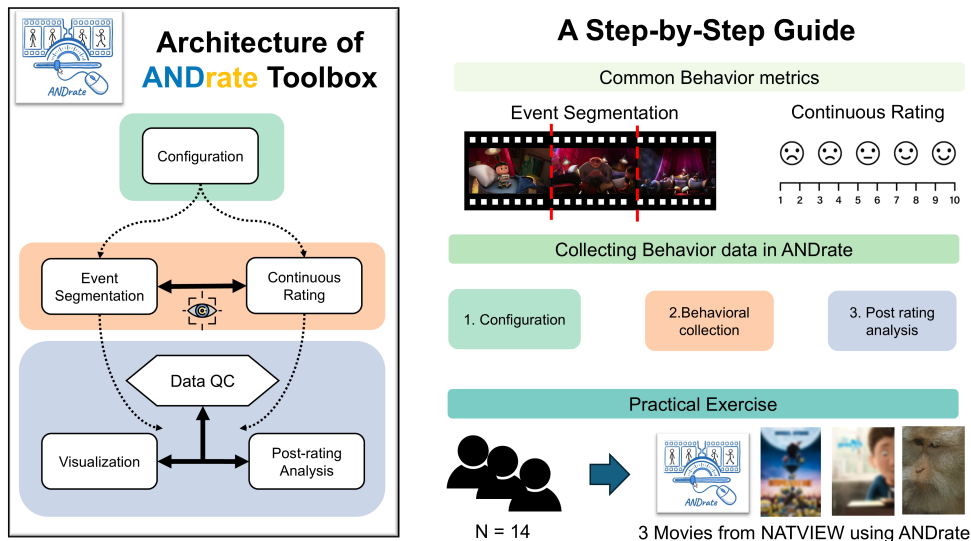


Figure 1: Overview of ANDrate Toolbox. Where it involves 3 major component, configuration, data collection including event segmentation and continuous rating, and data analysis including data QC, visualization and post-rating analysis. Moreover, web-based eye tracking was also implemented into ANDrate. To facilitate the usage of ANDrate, we outlined a step-by-step guide, introducing common behavior metrics, how to collect data in ANDrate, and finally provide an example using the toolbox

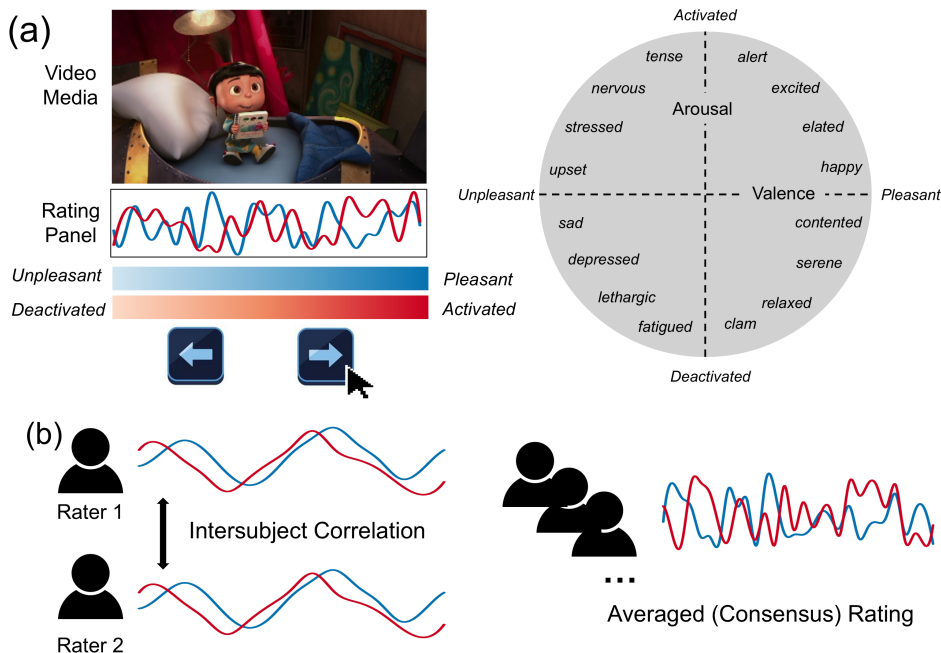


Figure 2: Summary of continuous rating task. 2a showing the a two-dimensional rating example where participants rate based on circumplex model of affect shown on the right of the figure. Specifically, as the video progresses, the participants would have to rate continuously based on Unpleasant-pleasant, and deactivated-activated continuum. 2b showing common analysis methods, including inter-subject correlation where the correlation coefficient between raters were calculated, often as a metric for synchronization of rating, as well as averaged rating across participants, often as a metric for consensus rating scores at group level.

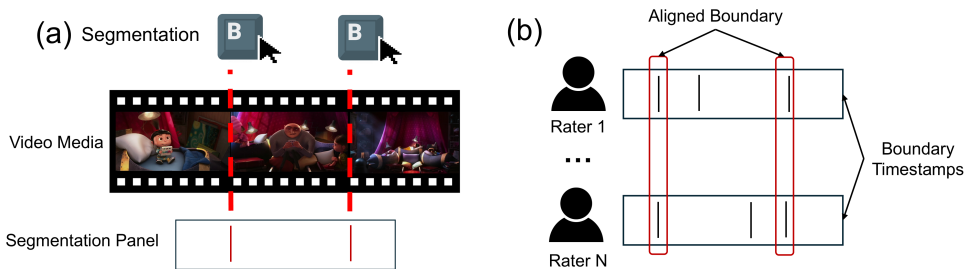


Figure 3: Summary of segmentation task. 3a showing the a example where participants press a button whenever they feel an event has ended, which the timestamp of these boundaries are marked on the panel. 2b showing one common analysis methods for discovering consensus boundaries, specifically, by using some alignment windows size and calculate the percentages of overlap across all participants.

STEP 1: Meta Information

Subject Prefix: e.g. dme, monkey1, p

Subject ID: 01

Save File Name: e.g. session_01

Upload Video / Audio: No file chosen

Select Target Media: Sample_Video.mp4

STEP 2: Global Task Settings

Segmentation Sampling Frequency: 1000.00 ms

Rating Sampling Frequency: 200 ms

X Axis Name & Range: Valence -5 5

Y Axis Name & Range: Arousal -5 5

Eye Tracking Settings

Calibration Accuracy Threshold: 60 %

Eye Tracking Module: ☐

Rating: Disable Y Axis: ☐

Data Storage Formats

Media Upload

Sampling Frequency

Rating Dimensions

Figure 4: ANDrate Configuration GUI. The interface can be divided into 5 sections, Data storage formats, Media Upload options, Sampling Frequency Settings, Rating Dimensions, and finally Eye tracking Settings

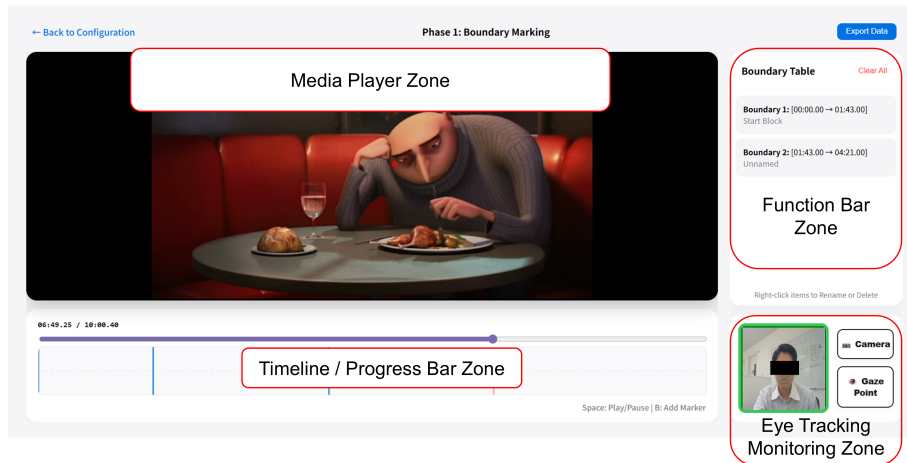


Figure 5: ANDrate Segmentation Task GUI. The interface can be divided into 4 Zones, Media Player Zone where stimuli play, Function Bar Zone where it records all the boundaries marked by participants, Timeline / Progress Bar Zone where it visualizes the timestamp of event boundaries, and finally Eye Tracking Monitoring Zone where it displays the status of eye gaze. The real-time gaze point is visualized as the red point on the screen.

Segmentation Data:

	Event Timestamp		Event Label
	A	B	C
1	Start	End	Label
2	0	48000	story telling
3	48000	86000	start to tell the story 1st time
4	86000	167000	story telling few times
5	167000	183000	end of telling story
6	183000	195000	go to sleep
7	195000	275000	start chatting with and old man
8	275000	291000	minions come
9	291000	300000	an object coming
10	300000	324000	the enjoy the tea together
11	324000	340000	an old woman comes
12	340000	355000	the woman slaps the man
13	355000	366000	the man brings the girls come out
14	366000	389000	girls ready to go away
15	389000	408000	the man chats with the old man again
16	408000	429000	the man dinner himself and see the minions clean the wall
17	429000	438000	the man ready to sleep and scared by a doll
18	438000	525000	many minions come again the man ready to fly to the space
19	525000	546000	another old man with yellow shirt and scared by a big object
20	546000	571000	someone fly to sky and meet with the man
21	571000	587000	the rocket fly to the space

Figure 6: Segmentation Task Data Output. The red column indicates the start and end timestamp for the event. And column in blue indicates the text label participants give to the event

Eye Tracking Data:

	Gaze Timestamp		Gaze coordinate		Area of Interest		
	A	B	C	D	E	F	G
1	Calibration_Accuracy	76					
2	Time(ms)	Gaze_X	Gaze_Y	On_Video	On_Funct	On_Time	Others
3	15	465	150	TRUE	FALSE	FALSE	FALSE
4	16	439	210	TRUE	FALSE	FALSE	FALSE
5	18	430	254	TRUE	FALSE	FALSE	FALSE
6	30	434	306	TRUE	FALSE	FALSE	FALSE
7	59	489	331	TRUE	FALSE	FALSE	FALSE
8	86	524	374	TRUE	FALSE	FALSE	FALSE
9	117	522	414	TRUE	FALSE	FALSE	FALSE
10	163	508	459	FALSE	FALSE	TRUE	FALSE
11	203	530	497	FALSE	FALSE	TRUE	FALSE
12	237	549	505	FALSE	FALSE	TRUE	FALSE
13	265	584	491	FALSE	FALSE	TRUE	FALSE
14	297	616	489	FALSE	FALSE	TRUE	FALSE
15	333	669	448	FALSE	FALSE	TRUE	FALSE
16	365	685	445	FALSE	FALSE	TRUE	FALSE
17	396	713	418	FALSE	FALSE	FALSE	TRUE
18	426	710	424	FALSE	FALSE	FALSE	TRUE
19	454	704	449	FALSE	FALSE	TRUE	FALSE
20	481	705	476	FALSE	FALSE	TRUE	FALSE
21	508	693	441	FALSE	FALSE	TRUE	FALSE
22	535	680	482	FALSE	FALSE	TRUE	FALSE
23	560	693	447	FALSE	FALSE	TRUE	FALSE
24	588	688	436	FALSE	FALSE	TRUE	FALSE
25	612	676	394	TRUE	FALSE	FALSE	FALSE
26	643	706	386	TRUE	FALSE	FALSE	FALSE
27	675	826	324	TRUE	FALSE	FALSE	FALSE
28	700	975	253	TRUE	FALSE	FALSE	FALSE
29	727	1104	189	TRUE	FALSE	FALSE	FALSE
30	753	1172	169	TRUE	FALSE	FALSE	FALSE

Figure 7: Eye Tracking Data Output. The purple row indicates the eye tracking Calibration Accuracy, the red column indicates the timestamp for each gaze point, the blue columns indicates the coordinates of the gaze point, and the green columns indicates the area of interest the gaze point belongs to

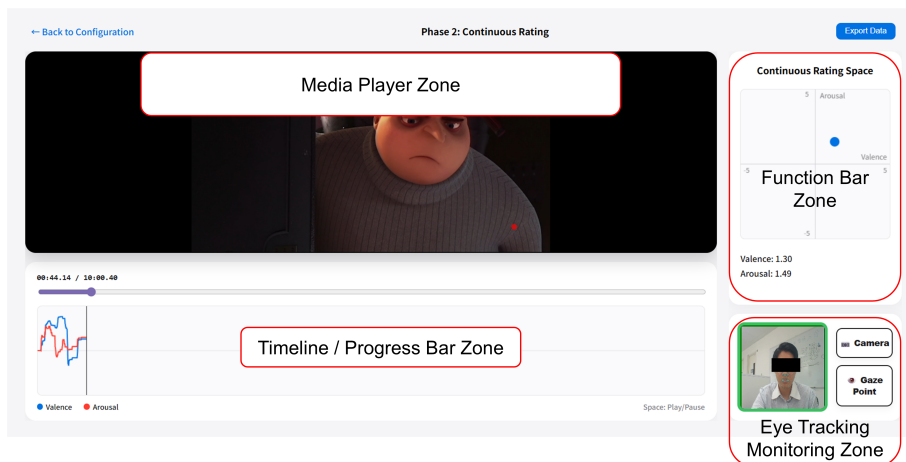


Figure 8: Continuous Rating Task GUI. The interface can be divided into 4 Zones, Media Player Zone where stimuli play, Function Bar Zone where it contains the continuous rating space in either one or two dimensions, Timeline / Progress Bar Zone where it visualize the rating scores, and finally Eye Tracking Monitoring Zone where it displays the status of eye gaze. The real-time gaze point is visualized as the red point on the screen

Rating Data:

Rating
Timestamp X and Y axis labels

	A	B	C
1	Time(ms)	Valence	Arousal
2	0	-0.012	0.02
3	200	-0.012	0.02
4	400	-0.012	0.02
5	600	-0.012	0.02
6	800	-0.012	0.02
7	1000	-0.012	0.02
8	1200	-0.012	0.02
9	1400	-0.012	0.02
10	1600	0.466	0.418
11	1800	1.661	1.932
12	2000	2.1	2.769
13	2200	2.219	2.888
14	2400	2.219	2.968
15	2600	2.299	3.048
16	2800	2.299	3.048
17	3000	2.299	3.048
18	3200	2.299	3.048
19	3400	2.299	3.048
20	3600	2.299	3.048
21	3800	2.339	3.207
22	4000	2.339	3.207
23	4200	2.339	3.207
24	4400	2.339	3.207
25	4600	2.378	3.247
26	4800	2.378	3.247
27	5000	2.378	3.247
28	5200	2.458	3.406
29	5400	2.458	3.406
30	5600	2.458	3.406

Figure 9: Continuous Rating Task Data Output. The red column indicates the timestamp for each rating score. The columns in blue indicate the rating scores.

ANDrate Tutorial for collecting behavioral data

Data Selection

Load Mode

Auto: Select data folder

Data Folder

SampleFile

Select Target Media

Sample_Video.mp4

Auto load will use folder: SampleFile

Consensus boundary window

4

seconds (±)

Min rater overlap

50

% of raters in cluster

Consensus Boundaries Options

Analysis Options

☒ 1. Eye-tracking QC
Per-participant attention-region statistics and calibration accuracy

☒ 2. Continuous Rating — ISC
Leave-one-out Pearson correlation per dimension

☒ 3. Continuous Rating — Group Average
Group mean ratings synced to video timeline

☒ 4. Segmentation — Consensus Boundaries
Cluster boundaries within ±window; label clouds for individual & group

☒ 5. Summary — Rating + Consensus Boundaries
Overlay group mean ratings with consensus segmentation boundaries

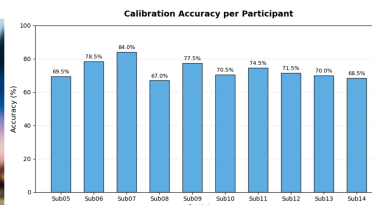
Analysis Panel

Run Selected Analyses

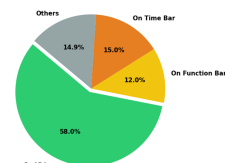
Figure 10: Post Rating Analysis GUI. The interface can be divided into 3 sections, Data Import Options, consensus boundaries options, and finally the analysis panel where analysis options could be selected

Despicable Me

Calibration
Accuracy: 73.2

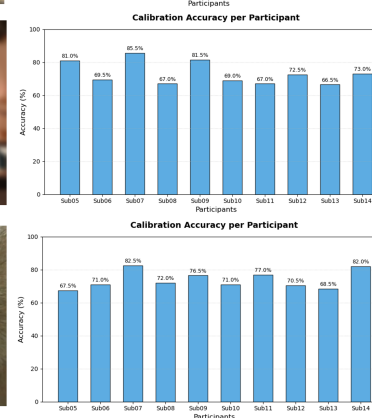


Attention Region Distribution (Group Mean)

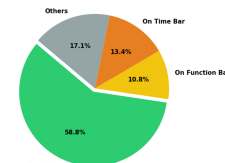


The Present

Calibration
Accuracy: 73.3

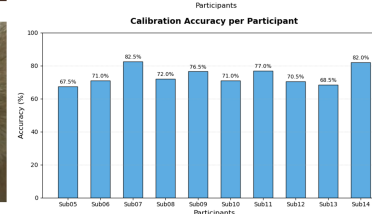
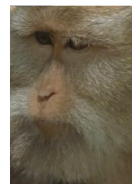


Attention Region Distribution (Group Mean)



Monkey Video

Calibration
Accuracy: 73.8



Attention Region Distribution (Group Mean)

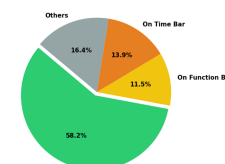


Figure 11: Summary of Eye Tracking Analysis. Specifically, the figure shows the calibration accuracy for each participants that contain eye tracking file in histogram as well as the distribution of attention at group level in pie chart, green: On Video, Yellow: On Function bar, Orange: On Time bar, grey: others.

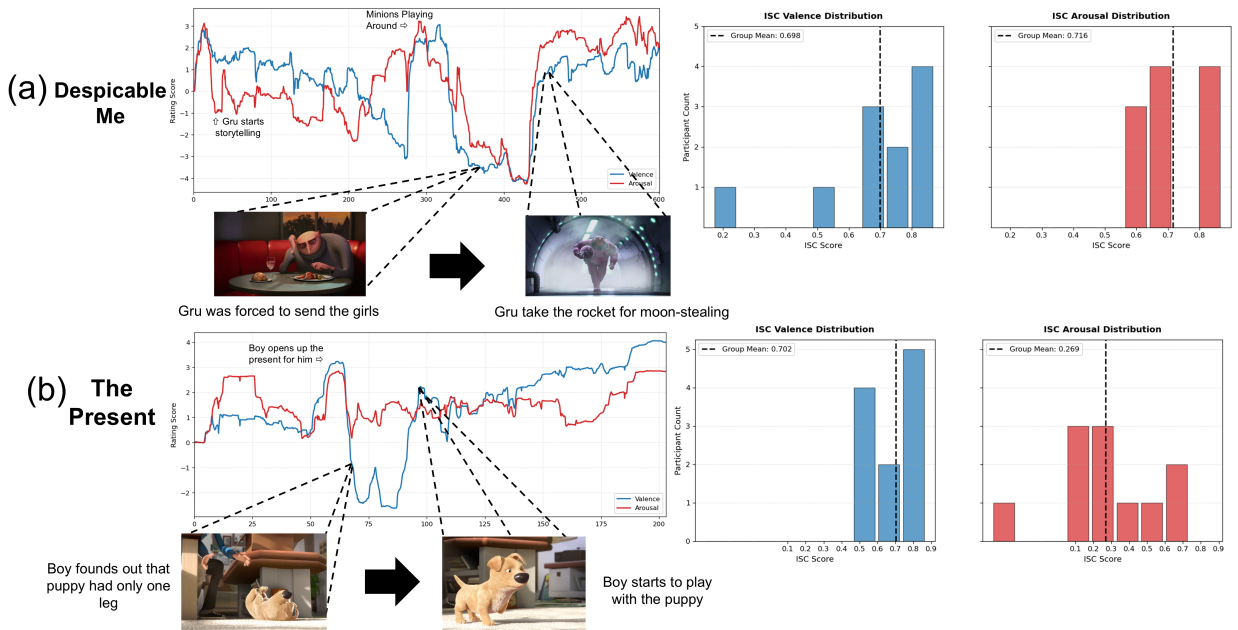


Figure 12: Summary for Continuous Rating Analysis for (a) Despicable Me and (b) The Present. The blue line indicates consensus rating for valence, and the red line indicates consensus rating for arousal. The results of inter-subject correlation are shown on the right, with black dashed line indicating the group mean of ISC scores.

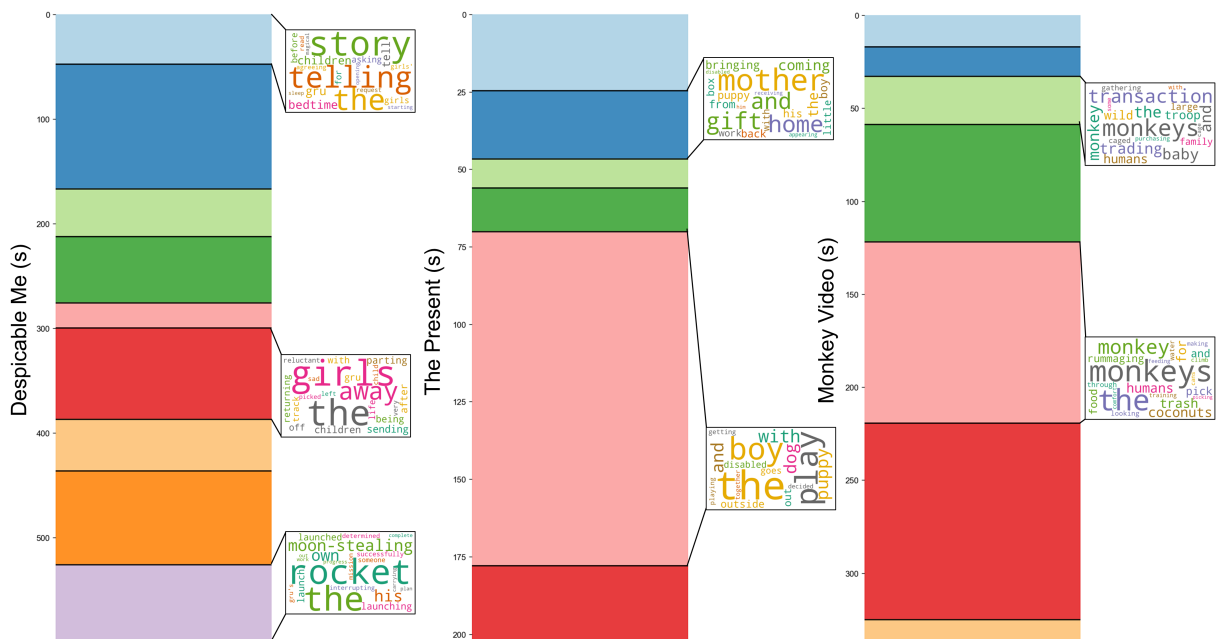


Figure 13: Summary for Boundary Segmentation Analysis. The y-axis shows the time in second. Each colored block indicates an consensus event with the corresponding word cloud besides to the event.

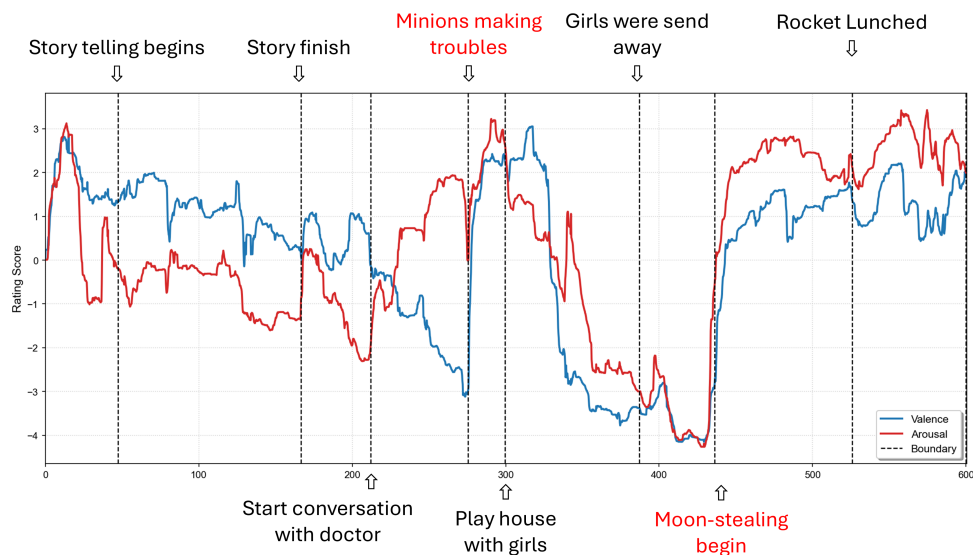


Figure 14: Relationship between boundaries and rating scores in the stimuli titled "despicable me". The dashed line indicates consensus boundaries identified at group level, while the blue and red line indicates consensus rating for valence and arousal identified at group level. The event boundaries often mark a shift in affective ratings. For example, when the minions are making troubles, or when the moon-stealing plan begins, the valence and arousal rise to a peak.